

Decoupled Image-to-Video Generation via Latent Flow: An Investigation into Embedding-Conditioned Rendering

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Abstract

001 *Image-to-video generation from a single static image and*
002 *an action label requires simultaneously understanding both*
003 *what is in the scene and how it should evolve over time.*
004 *We investigate a decoupled framework inspired by the Joint-*
005 *Embedding Predictive Architecture (JEPA): a causal Trans-*
006 *former (Flow Model) predicts the trajectory of CLIP em-*
007 *beddings in a compact 512-dimensional semantic space,*
008 *and a conditioned UNet (Renderer) converts each predicted*
009 *embedding back to a pixel frame. We conduct a system-*
010 *atic set of experiments across four Renderer architectures—*
011 *FiLM conditioning, cross-attention to CLIP patch tokens,*
012 *contrastive conditioning loss, and optical flow prediction—*
013 *and find that each variant collapses to a mode where the em-*
014 *bedding condition is effectively ignored and the output is a*
015 *near-identity transformation of the input frame. We present*
016 *a detailed failure analysis identifying the root causes: van-*
017 *ishing embedding motion signal between adjacent frames,*
018 *a training objective fundamentally misaligned with condi-*
019 *tioning sensitivity, and the inherent difficulty of encoding*
020 *spatial motion into a single global feature vector. Despite*
021 *the negative result, our analysis highlights key design chal-*
022 *lenges for embedding-conditioned video synthesis and sug-*
023 *gests concrete alternative approaches.*

024 1. Introduction

025 Image-to-video (I2V) generation aims to produce a tempo-
026 rally coherent video sequence from a single static frame.
027 When conditioned on an action label (e.g., “walking”), the
028 generated video should depict natural motion consistent
029 with that action. This task is relevant to content creation,
030 video editing, and synthetic data augmentation for action
031 recognition systems.

032 Recent large-scale diffusion models such as Stable Video
033 Diffusion and Wan2.2 have demonstrated impressive results
034 by inflating image diffusion architectures into the tempo-
035 ral domain. However, these end-to-end approaches incur

substantial computational costs and remain prone to motion
flickering and identity drift.

Inspired by the JEPA framework, we investigate an al-
ternative: decoupling the problem into a **Flow Model** that
predicts motion in a semantic embedding space and a **Ren-**
derer that converts semantic predictions back to pixels. The
key hypothesis is that motion dynamics are easier to model
in a compact, abstract embedding space than in raw pix-
els. In an ideal scenario, the Flow Model would learn high-
level action dynamics (“a walking person alternates legs”),
and the Renderer would faithfully translate these semantic
blueprints into frames.

We implement this framework using frozen CLIP ViT-
B/32 encoders, a causal Transformer for embedding trajec-
tory prediction, and a UNet-based Renderer with condition-
ing injected via FiLM, cross-attention, or explicit flow pre-
diction heads. We systematically ablate the conditioning
mechanism and loss function across four Renderer variants.

Our key finding is negative but instructive: across all
four variants, the Renderer learns to ignore the embedding
condition and produces near-identical outputs regardless of
input conditioning. The root cause is not any single archi-
tectural choice but a fundamental tension between (1) the
minimal variation in CLIP embeddings between adjacent
video frames, and (2) loss functions that penalize deviation
from a nearly-static ground truth.

We present a detailed failure analysis quantifying the
embedding signal strength, the collapse behavior across
architectures, and the training dynamics. We believe
these findings provide useful guidance for future work on
embedding-conditioned generative models.

2. Related Work

2.1. Image-to-Video Generation

Early video generation methods employed recurrent net-
works or generative adversarial networks. More recently,
diffusion-based approaches have become standard. Stable
Video Diffusion fine-tunes a pre-trained image diffusion
model on video data by adding temporal layers. Wan2.2 fol-

074 lows a similar approach at larger scale. These models gener-
075 ally operate in a VAE latent space, simultaneously modeling
076 spatial structure and temporal dynamics in one pipeline. We
077 experiment with a fundamentally different architecture that
078 attempts to factor these responsibilities across two modules.

079 2.2. Latent Predictive Architectures

080 The JEPA framework proposes that predictive learning
081 should operate in representation space rather than input
082 space. V-JEPA applies this principle to self-supervised
083 video learning by predicting masked spatio-temporal fea-
084 tures. Our Flow Model follows the same intuition: pre-
085 dict the trajectory of CLIP embeddings rather than raw pix-
086 els. Where our work differs is the downstream rendering
087 step—converting predicted embeddings back to pixels—
088 which proves to be the critical bottleneck.

089 2.3. Conditioned Image Generation

090 Feature-wise Linear Modulation (FiLM) and cross-
091 attention are standard techniques for conditioning image
092 generation on external information. FiLM applies channel-
093 wise affine transformations $\gamma(c) \odot x + \beta(c)$ predicted from
094 a conditioning vector c , and has been widely used in vi-
095 sual question answering, style transfer, and video predic-
096 tion. Cross-attention, popularized by the Transformer ar-
097 chitecture, allows each spatial position in a feature map to
098 attend to a set of conditioning tokens. Our experiments test
099 both mechanisms independently and in combination, and
100 find that neither is sufficient when the conditioning signal
101 (adjacent-frame CLIP embeddings) varies by less than 3%
102 in cosine distance.

103 2.4. Optical Flow and Warping

104 Frame interpolation and video prediction often employ ex-
105 plicit optical flow prediction followed by warping. Methods
106 such as RAFT achieve state-of-the-art optical flow estima-
107 tion through iterative refinement, while simpler approaches
108 use Farneback or TV-L1 flow as a supervision signal. Our
109 final Renderer variant adopts a flow-based approach: a
110 UNet backbone predicts a dense flow field conditioned on
111 CLIP embeddings, and the previous frame is warped via
112 differentiable bilinear sampling. We use Farneback optical
113 flow as pseudo-supervision and apply total variation regu-
114 larization to encourage spatial smoothness.

115 2.5. Action-Conditioned Video Generation

116 Motion generation conditioned on action labels has been
117 studied in controlled settings such as video game environ-
118 ments and simplified human motion benchmarks. Text-to-
119 video models such as Make-A-Video and Imagen Video
120 use large language model embeddings for open-vocabulary
121 conditioning at scale. Our setting is intermediate: we use

closed-set action labels encoded through CLIP text embed-
dings, providing a clean semantic signal in the same embed-
ding space as the visual features.

3. Data

We use the HMDB51 dataset, containing 6,754 video
clips across 51 human action categories such as “walking,”
“drinking,” “brushing hair,” and “climbing stairs.” Videos
average approximately 3 seconds in duration at 25–30 fps.
The dataset is split into training (70%) and test (30%) sets
following the first official split.

3.1. Preprocessing

For each video, we sample 17 consecutive frames ($n = 16$
prediction steps) at a stride of 2, and resize all frames to
 224×224 pixels. CLIP embeddings (CLS token + 49 patch
tokens) are pre-extracted for the entire dataset (242MB) to
avoid redundant forward passes during Flow Model train-
ing. The Renderer samples random frame pairs (I_{t-1}, I_t)
from each video with a gap of 1 frame.

3.2. Embedding Analysis

A key diagnostic we perform early is measuring the varia-
tion in CLIP embeddings across frames. Table 1 summa-
rizes pairwise cosine similarities.

Table 1. CLIP ViT-B/32 embedding cosine similarity. Same-video
frames are highly similar ($1 - \cos = 0.024$ for adjacent frames),
limiting the information available for frame-to-frame motion pre-
diction.

Comparison	Mean cos
Same video, distant frames ($\Delta t \approx 60$)	0.862
Same video, adjacent frames ($\Delta t = 1$)	0.976
Different videos, same action class	0.664
Different action classes	0.564

The critical observation is that adjacent-frame embed-
dings differ by only $1 - \cos \approx 0.024$ on average. This means
the embedding trajectory that the Flow Model must predict,
and that the Renderer must use as a conditioning signal,
contains very little per-frame variation. Any learning algo-
rithm must amplify this tiny signal by orders of magnitude
to produce meaningful pixel-space changes.

4. Methods

Our framework consists of three frozen encoders and two
trainable modules. Figure 1 illustrates the overall pipeline.

4.1. Encoders

Image Encoder. Frozen CLIP ViT-B/32. We extract both
the 512-dim CLS embedding E_t and the 49 spatial patch to-



Figure 1. Overall pipeline. Frozen CLIP encoders produce image embeddings E_t and label embeddings L . The Flow Model (causal Transformer) predicts $(\hat{E}_1, \dots, \hat{E}_n)$ from (L, E_0) . The Renderer (UNet with various conditioning mechanisms) produces each frame I_t from I_{t-1} conditioned on $\hat{E}_t$ and L .

kens (7×7 grid, 512-dim each) from the final vision transformer layer.

Label Encoder. Frozen CLIP text encoder. Action labels are converted from directory names to natural text (e.g., “brush hair”) and encoded to 512-dim semantic embeddings L .

4.2. Flow Model

A 4-layer causal Transformer encoder (12.9M parameters) with 8 attention heads and feedforward dimension 2048. The input sequence is $[L, E_0, E_1, \dots, E_{n-1}]$, and the model is trained with shifted teacher forcing: position t predicts $\hat{E}_t$ from $(L, E_0, \dots, E_{t-1})$.

We compare two loss functions:

- **MSE:** $\mathcal{L} = \frac{1}{n} \sum_t \|\hat{E}_t - E_t\|_2^2$
- **Cosine:** $\mathcal{L} = \frac{1}{n} \sum_t (1 - \cos(\hat{E}_t, E_t))$

4.3. Renderer

The Renderer is a UNet (4 down/up blocks, 40–43M parameters depending on variant) that takes I_{t-1} as input and produces an output conditioned on $\hat{E}_t$, L , and patch tokens. We experiment with four variants:

1. **FiLM (baseline):** channel-wise Feature-wise Linear Modulation from a condition vector $[\hat{E}_t; L] \in \mathbb{R}^{1024}$, applied at each residual block. Output: pixel residual ΔI_t . Loss: $L1(\Delta I_t, I_t - I_{t-1})$.
2. **FiLM + Cross-Attention:** FiLM conditioning plus multi-head cross-attention from UNet feature maps to CLIP patch tokens at the bottleneck and first two upsampling blocks. Output: pixel residual ΔI_t . Loss: L1.
3. **FiLM + Cross-Attention + Contrastive:** same as variant 2, plus a contrastive conditioning loss: for a batch, we compute $\mathcal{L}_{\text{cont}} = \|\cos(\Delta I_{\text{real}}, \Delta I_{\text{fake}}) - \cos(E_{\text{real}}, E_{\text{fake}})\|_2^2$, where fake conditions are shuffled embeddings from other videos in the batch. This explicitly penalizes the Renderer for producing identical outputs from different conditions.
4. **Flow Prediction:** same UNet backbone with FiLM + cross-attention, but the output head predicts a dense optical flow field $\in \mathbb{R}^{2 \times 224 \times 224}$. The previous frame is warped via the predicted flow: $I_t = \text{warp}(I_{t-1}, \text{flow})$.

$$\text{Loss: } L1(I_t^{\text{warped}}, I_t^{\text{gt}}) + \lambda_{\text{TV}} \cdot \text{TV}(\text{flow}) + \lambda_{\text{FB}} \cdot L1(\text{flow}, \text{Farneback}).$$

4.4. Training Details

All models are trained with AdamW ($lr = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}) and a cosine annealing learning rate schedule over 100 epochs. We use a batch size of 8 with gradient accumulation over 2 steps (effective batch size 16). Gradient clipping with maximum norm 1.0 is applied. All experiments were conducted on a single NVIDIA RTX 4060 Laptop GPU (8GB VRAM). Complete training of the Flow Model takes approximately 4 hours; each Renderer variant takes approximately 7 hours.

Flow Model training data pipeline. Because passing CUDA models through PyTorch DataLoader workers is not supported on Windows, we pre-extract CLIP embeddings for all 6,754 videos (17 frames each) and save them to disk as a single 242MB file. The Flow Model training loop loads pre-extracted embeddings directly, requiring no CLIP forward passes during training. This eliminates the DataLoader CUDA issue and accelerates training by approximately $3 \times$.

Renderer training data pipeline. The Renderer uses HMDB51FrameDataset, which loads raw frames from disk (via OpenCV) and performs image transformations in the DataLoader workers. CLIP encoding happens in the main training loop where CUDA is available. Farneback optical flow is computed on CPU using OpenCV, adding approximately 30% overhead per batch.

4.5. Implementation Notes

Shifted teacher forcing. An earlier version of the Flow Model used “zero-placeholder” inputs where the model received $[L, E_0, \mathbf{0}, \dots, \mathbf{0}]$ and was expected to predict all future embeddings in a single forward pass. This created a train-inference mismatch: at inference time, the model sees its own predictions rather than zeros. Shifted teacher forcing, where the model receives the real embedding sequence and predicts one step ahead at each position, aligns training and inference exactly. We found this improved embedding trajectory quality significantly (from 2.0% to 82.5% of GT motion, combined with the cosine loss change).

Multi-scale flow prediction. In the flow prediction Renderer variant, we considered predicting flow at multiple decoder resolutions and combining them via coarse-to-fine refinement (similar to RAFT). However, initial experiments showed that even a single-scale flow head struggled to produce consistent non-zero outputs, so we deferred multi-scale investigation to future work.

5. Experiments

5.1. Evaluation Protocol

We evaluate using the following metrics:

- **Embedding Trajectory MSE:** MSE between predicted and ground-truth CLIP embedding sequences.
- **Inter-frame 1-cos Distance:** $1 - \cos(\hat{E}_t, \hat{E}_{t+1})$, measuring how much “motion” the predicted embeddings contain.
- **Conditioning Sensitivity:** $\frac{\|\text{Renderer}(I, E_t) - \text{Renderer}(I, E_0)\|_1}{\|\text{Renderer}(I, E_0)\|_1 + \epsilon}$, measuring whether changing the embedding changes the Renderer output.

5.2. Flow Model Results

Table 2 summarizes the Flow Model results.

Table 2. Flow Model performance with MSE vs. cosine similarity loss. Cosine loss prevents embedding collapse while maintaining prediction accuracy.

Metric	MSE Loss	Cosine Loss
Train loss (final)	0.0116	0.0264
Inter-frame 1-cos (GT)	0.0290	0.0290
Inter-frame 1-cos (pred)	0.0006	0.0239
% of GT motion preserved	2.0%	82.5%

The MSE-trained Flow Model achieves low MSE (0.0116) but collapses to predicting nearly identical embeddings at every timestep ($1 - \cos \approx 0.0006$). Cosine loss resolves this: predicted embeddings retain 82.5% of the ground-truth motion variation while maintaining comparable embedding quality (MSE between predicted and GT embeddings actually improves in terms of cosine distance).

5.3. Renderer Results

Table 3 presents the core negative result: across all four architectures, the Renderer learns to ignore the embedding condition.

For all variants, the conditioning sensitivity ratio is approximately 1.0, meaning the magnitude of the Renderer’s output does not meaningfully change when the conditioning embedding changes. In variants 1–3 (residual prediction), the absolute output magnitude converges to ≈ 0.005 —essentially near-zero. In variant 4 (flow prediction), the pre-

Table 3. Renderer conditioning sensitivity across four architecture variants. A ratio of ~ 1.0 indicates the model produces identical outputs regardless of the embedding condition.

Renderer Variant	Sensitivity Ratio
1. FiLM + L1	1.03x
2. FiLM + Cross-Attn + L1	0.98x
3. FiLM + Cross-Attn + L1 + Contrastive	0.98x
4. FiLM + Cross-Attn + Flow Pred + Rec + TV + FB	1.01x

dicted flow field converges back to zero after initial non-zero activity from random initialization.

5.4. Diagnostic Experiments

To isolate the failure mode, we conduct several diagnostic tests:

Embedding distance test. We feed the Renderer embeddings from frames at varying temporal distances ($[t, t + 1, t + 4, t + 8, t + 16]$) and measure the output magnitude. Even with frame $t + 16$ ($1 - \cos = 0.31$, a meaningful change), the output magnitude remains unchanged from the $t + 1$ case ($1 - \cos = 0.03$). This confirms that the model is not merely limited by the weak signal in adjacent frames—it has learned to ignore the embedding entirely.

Farneback flow as pseudo-target. We compute Farneback optical flow between adjacent frames (mean magnitude 0.39 px, max 2.84 px) and use it as a supervised target. The flow head initially produces non-zero flow but converges back to zero as the reconstruction and smoothness losses dominate.

CLIP feature discrimination. We verify that CLIP embeddings *do* contain frame-to-frame information: the cosine similarity gap between same-video adjacent frames (0.976) and different videos of the same class (0.664) is 0.312—a substantial difference. The information exists, but translating it from a 512-dim global vector into spatially meaningful pixel changes proves fundamentally difficult with our architectures.

6. Discussion: Why Did It Fail?

We identify the following contributing factors, ordered from most fundamental to most specific.

6.1. Primary Cause: CLIP Embeddings Lack the Necessary Information

We believe this is the root cause from which all other issues derive. CLIP ViT-B/32 is trained with a contrastive image-text matching objective: given an image and a text description, maximize the cosine similarity of their embeddings. This objective encourages the model to extract *semantic content*—“what is in this image”—and compresses it into a compact 512-dimensional vector.

312	What CLIP’s training objective does <i>not</i> require is pre-	360
313	serving the kind of information needed for frame-to-frame	361
314	rendering:	362
315	• Spatial configuration: The precise 2D coordinates of	363
316	body parts, objects, and their relative positions.	364
317	• Pose and articulation: The joint angles of limbs, the ori-	365
318	entation of a head, the bend of a knee.	366
319	• Dense correspondence: Which pixel in frame t corre-	367
320	sponds to which pixel in frame $t + 1$.	368
321		369
322	A CLIP embedding can tell you “this is a person walk-	370
323	ing” with high confidence, but it cannot tell you “the right	371
324	leg is currently forward at a 30-degree angle, positioned at	372
325	pixel coordinates (120, 180).” The latter is precisely the in-	
326	formation a Renderer needs to predict what the next frame	
327	should look like.	
328	This explains the asymmetry in our results: the Flow	
329	Model succeeded because predicting an embedding trajec-	
330	tory only requires matching the statistical distribution of	
331	CLIP embeddings—a task CLIP embeddings contain suf-	
332	ficient information for. The Renderer failed because de-	
333	coding an embedding back to a specific spatial configura-	
334	tion of pixels requires information that was discarded dur-	
335	ing CLIP’s encoding process.	
336	Empirically, we confirmed this information bottleneck:	
337	adjacent frames of the same video have CLIP embeddings	
338	with cosine similarity ≈ 0.976 ($1 - \cos \approx 0.024$). This	
339	tiny difference must encode the entire motion transforma-	
340	tion from one frame to the next—a compression ratio far	
341	exceeding what any decoder architecture can plausibly in-	
342	vert.	
343		
344	6.2. Contributing Factor: Loss Function Misalign-	
345	ment	
346	All our loss functions (L1 on residuals, L1 on reconstruc-	
347	tion after warping) have an optimal trivial solution near the	
348	identity transformation: $\Delta I_t \approx 0$ and $\text{flow} \approx 0$. When the	
349	conditioning signal carries almost no usable information (as	
	argued above), identity mapping becomes the rational opti-	
	mum.	
350	6.3. Contributing Factor: FiLM Cannot Express	
351	Spatial Guidance	
352	Feature-wise Linear Modulation applies channel-wise scal-	
353	ing and shifting uniform across all spatial positions. Even if	
354	the embedding <i>did</i> contain spatial information, FiLM lacks	
355	the mechanism to route it to the correct spatial location.	
356	Cross-attention to CLIP patch tokens partially addresses	
357	this routing problem, but the patch tokens themselves still	
358	lack dense correspondence information due to the primary	
359	cause above.	
	6.4. Limitations of Our Study	
	We acknowledge several limitations: (1) we only explored	
	CLIP ViT-B/32 as the embedding backbone—larger ViT	
	variants or dense-prediction models like DINOv2 may have	
	different properties; (2) we did not attempt joint end-to-end	
	training of the Flow Model and Renderer, which could al-	
	low the Flow Model to learn embeddings specifically opti-	
	mized for rendering; (3) we used a relatively small dataset	
	(6.7k videos), and the collapse behavior may differ at larger	
	scale where more diverse motion patterns provide a stronger	
	learning signal; (4) HMDB51 videos are constrained to spe-	
	cific human action categories, and the learned dynamics	
	may not generalize to more open-ended video generation.	
	6.5. What We Learned	
	Despite the negative result, this investigation yielded several	
	lessons that may benefit future work in this area:	
	Embedding choice is critical. The choice of CLIP ViT-	
	B/32 as our embedding backbone was motivated by its joint	
	vision-language space and compact dimensionality. How-	
	ever, CLIP embeddings are optimized for semantic dis-	
	crimination, not spatial correspondence. For a decoupled	
	I2V pipeline to succeed, the embedding space must encode	
	both <i>what</i> is in the scene and <i>where</i> things are spatially—	
	properties that CLIP’s contrastive training objective does	
	not guarantee.	
	Loss functions define the optimization landscape. We	
	observed that all tested pixel-space loss functions (L1, L1	
	+ TV, L1 + Farneback) contain a near-identity trivial so-	
	lution ($\Delta I \approx 0$, $\text{flow} \approx 0$). This is not specific to our	
	architecture—any model trained with a pixel reconstruction	
	loss on adjacent-frame pairs faces the same challenge. Loss	
	functions that operate in a learned feature space (perceptual	
	loss, adversarial loss) may provide a stronger signal for mo-	
	tion learning.	
	The JEPA philosophy needs a rendering pathway.	
	JEPA’s core insight—predict in representation space, not	
	input space—proved effective for our Flow Model (82.5%	
	of GT motion). But the companion challenge of <i>decod-</i>	
	<i>ing</i> those predictions back to the pixel domain is equally	
	hard and received comparatively little attention in our im-	
	plementation. The rendering problem deserves its own care-	
	ful treatment rather than being treated as an architectural	
	afterthought.	
	6.6. Future Work	
	Several promising directions emerge from this analysis.	
	Dense spatial embeddings. Replacing CLIP with	
	a model trained for dense visual correspondence—such	
	as DINOv2, Stable Diffusion’s VAE latents, or a self-	
	supervised optical flow network—could provide embed-	
	dings that naturally encode spatial configuration. A tra-	
	jectory in DINOv2 patch feature space would carry richer	

spatial information than a trajectory in CLIP CLS space.

Joint end-to-end training. Training the Flow Model and Renderer jointly, with gradients flowing from the rendering loss back through the Flow Model, would allow the embedding space to adapt to the needs of the Renderer. The Flow Model would no longer be constrained to match CLIP’s embedding structure and could learn features specifically optimized for frame rendering.

Iterative refinement. Rather than predicting the full frame transformation in a single forward pass, a multi-step refinement process (similar to diffusion models or recurrent refinement) could gradually build up the output from an initial near-identity estimate. This would allow the model to learn motion in stages—coarse structure first, then fine details.

Adversarial and perceptual losses. GAN-based losses or CLIP-based perceptual losses directly penalize lack of motion and unrealistic static output, potentially breaking the near-identity local minimum that trapped all of our L1-trained models.

Explicit motion field supervision. Using a pre-trained optical flow network (e.g., RAFT) instead of Farneback as a supervision target could provide a cleaner learning signal. Combined with a rendering loss, this could guide the model toward physically plausible motion while maintaining visual fidelity.

7. Conclusion

We set out to build a decoupled image-to-video generation system in which a causal Transformer predicts action-conditioned embedding trajectories and a conditioned UNet renders those trajectories to pixels. The Flow Model component was partially successful: with cosine similarity loss it learned to predict embedding trajectories preserving 82.5% of ground-truth motion variation, while MSE loss led to complete collapse. However, the Renderer failed across all four conditioning architectures tested: FiLM, cross-attention, contrastive conditioning, and optical flow prediction.

Our analysis suggests the primary cause is a fundamental information bottleneck: CLIP ViT-B/32 embeddings are trained for semantic image-text alignment and do not preserve the spatial configuration, pose, and correspondence information necessary for pixel-level frame rendering. Adjacent frames differ by less than 3% in CLIP embedding cosine distance, an information compression ratio that is implausible for any decoder architecture to invert. All other observed problems—loss function collapse to identity, FiLM’s inability to express spatial motion—can be understood as consequences of this root cause.

Future directions suggested by this analysis include: (1) replacing CLIP with a dense visual representation trained for spatial correspondence, such as DINOv2 patch

features, Stable Diffusion VAE latents, or self-supervised optical flow features; (2) training the Flow Model and Renderer jointly end-to-end, so the embedding space can adapt to the specific needs of rendering rather than being constrained by a frozen pre-trained encoder; (3) adopting losses that explicitly penalize lack of motion, such as adversarial or perceptual losses, to break the near-identity local minimum; (4) incorporating explicit motion representations such as optical flow or keypoint trajectories as intermediate supervision targets rather than relying solely on the embedding to carry all motion information.

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